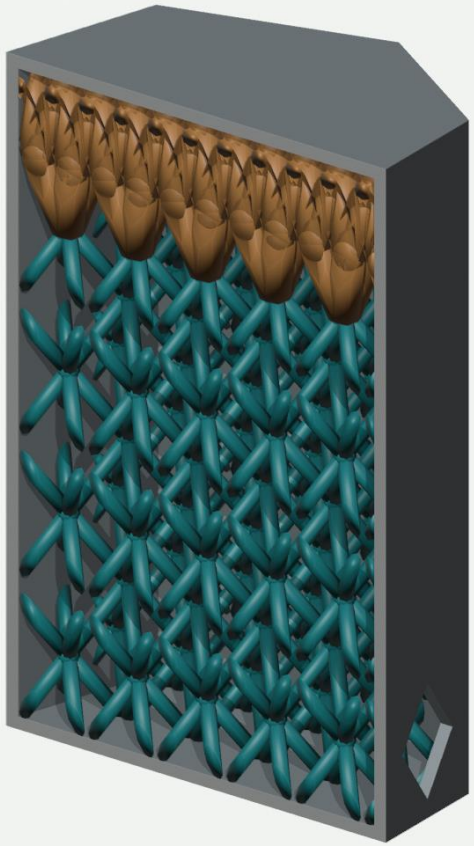
Volume to strut-lattice structures

Lattice filling for volume supports with an optional cover

Replace a solid interior with nodes and struts. Export the result as STEP/STP or STL.

Lattice filling can reduce material use. Production-time savings depend on the printing process.

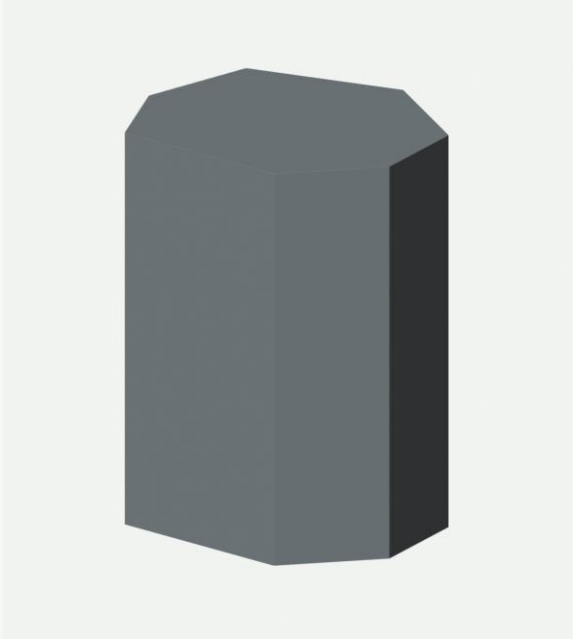


volume-to-strut-lattice-stp-review
volume-to-strut-lattice-stl-review

Volume transformation workflow

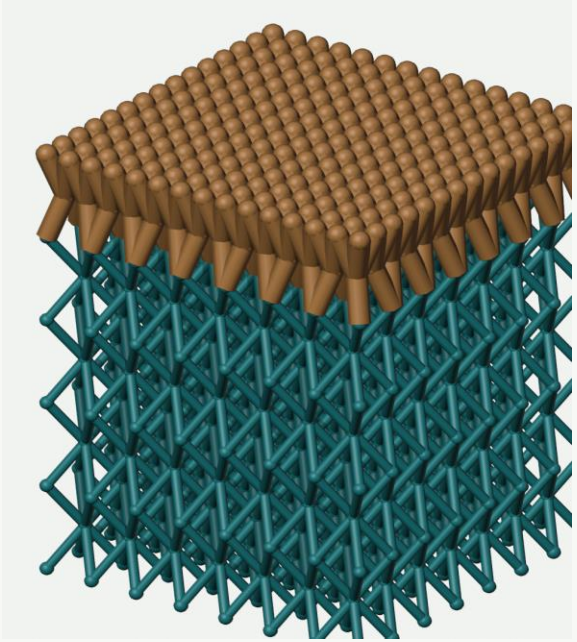
Review settings first. The STEP/STP solid then defines the volume for the strut lattice.

1. Source volume



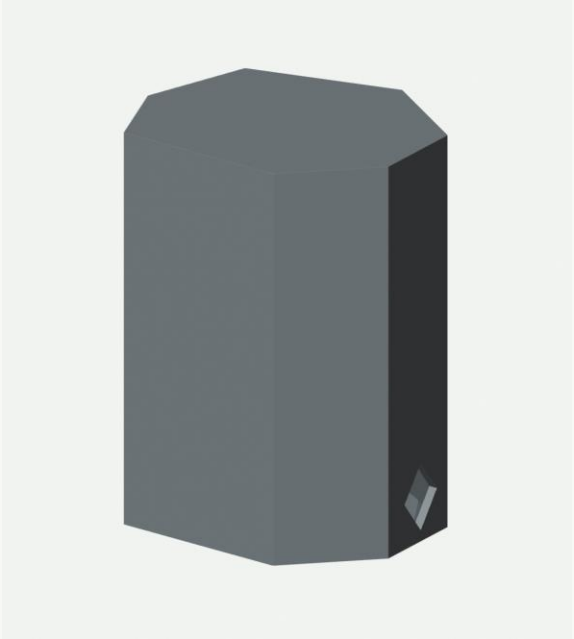
Irregular polygonal volume

2. Repeated lattice



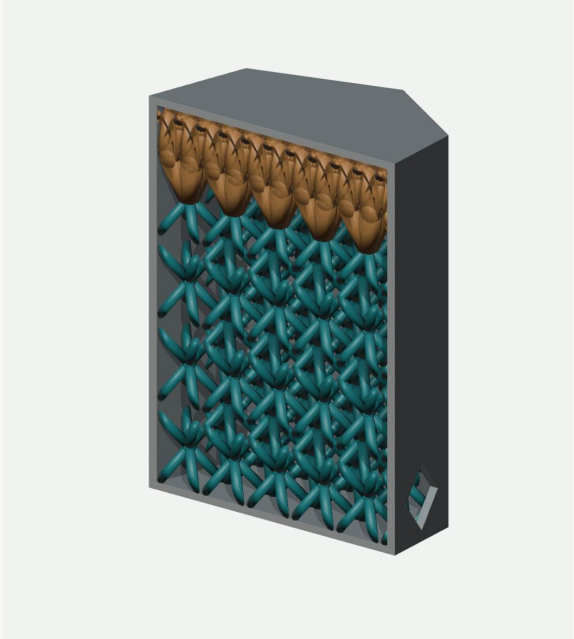
Fixed cell and strut sizes

3. Finished part



Covered example: 2 mm wall

4. Finished part in section



Local upper layers and opening

Choose STEP/STP CAD output or STL mesh output. Cover and powder opening are optional.

Editable lattice geometry parameters

Shared defaults for the STEP/STP and STL skills. Dimensions in millimetres.

Parameter	Default	What it controls
Lower cell pitch	12 mm	Horizontal spacing of lower nodes
Vertical layer spacing	11.1 mm	Regular spacing and preferred local upper drop
Lower node / strut diameter	3.3 / 2.4 mm	Junction size / lower member thickness
Connections per node	Up to 4	Maximum selected connections; geometry may limit it
Minimum strut angle	50°	Inclination above XY; selection adds a fixed 0.5°
Top contact pitch	6 mm	XY spacing of local upper contacts
Top node diameter	5 mm	Size of local upper junctions
Top strut diameter	5 mm	Constant diameter of top AND transition struts
Upper-surface threshold	Below 45°	Selects local planar / curved roof and ledge regions

Fixed: world +Z build direction and two local upper node levels. Local drops may shorten or shift.

Cover, opening and output parameters

The cover is optional. Opening controls are inactive when thickness is 0 mm.

Parameter	Default	What it controls
Cover thickness	2 mm	Inward skin; 0 mm creates lattice only
Opening width × height	12 × 18 mm	Diamond dimensions in the chosen side face
Opening side and centre X/Y/Z	Per body	Placement on a planar vertical side
Opening roof angle / clearance	50° / 6 mm	Minimum roof angle / extra inward exclusion depth
STL mesh deflection	0.03 mm	Mesh approximation; STL version only
Surface sampling / CAD tolerance	0.1 / 0.01 mm	Surface probes / numerical geometry checks
XY orientation hint	[1, 0, 0]	Rotation of the lower grid about +Z
Member budget	20,000	Computational guard for member generation

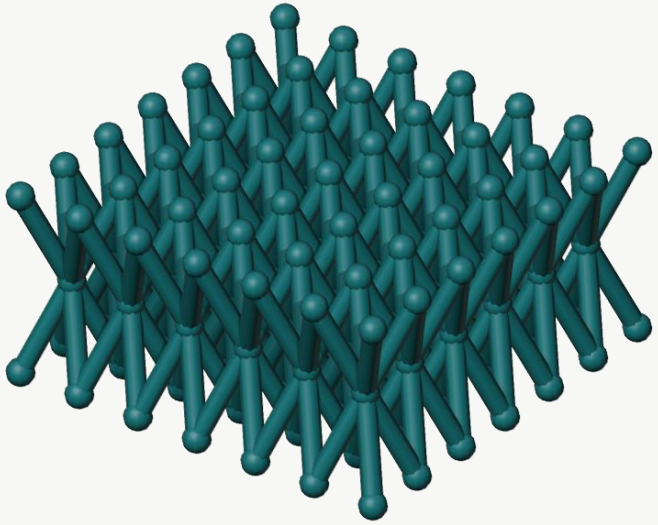
Job settings: input, work and output paths; model name; descriptive metadata such as alloy.

STEP/STP preserves CAD surfaces. STL exports a mesh. Both retain intersecting closed members.

Lower cell pitch

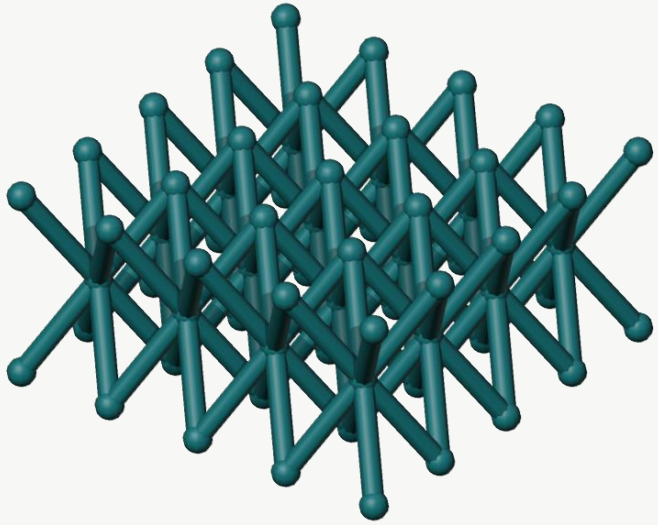
Same 48 × 48 mm node-center footprint. Node diameter 3.3 mm, strut diameter 2.4 mm.

8 mm



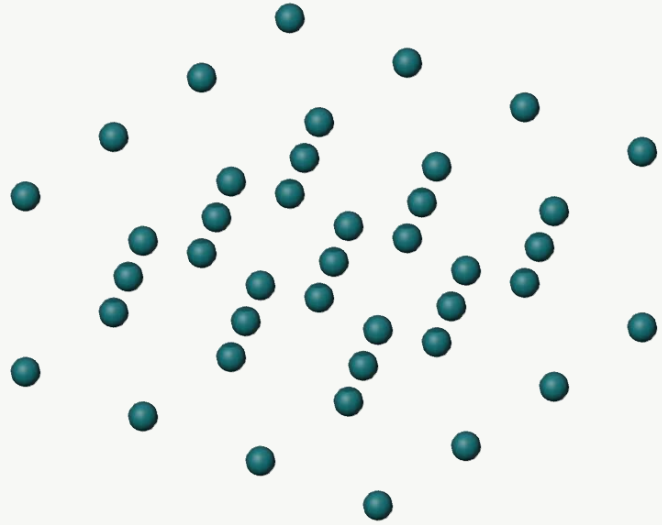
Closer nodes, steeper struts

12 mm standard



52.6° nearest connections

16 mm



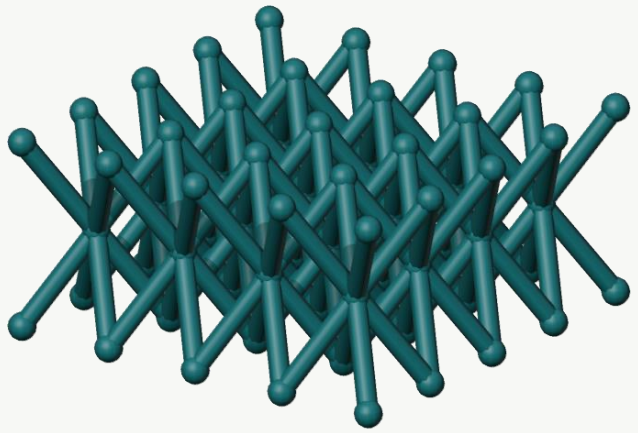
No eligible lower connections

Rise stays at 11.1 mm. At 16 mm pitch, the dots are candidates; unused nodes are omitted in final output.

Layer rise and pitch work together

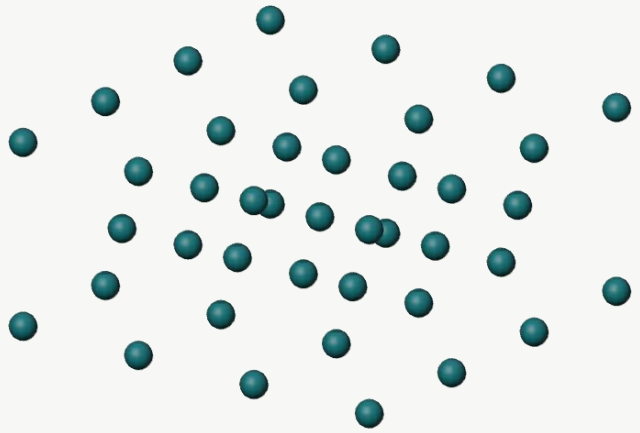
The lower grid uses the requested spacing. These simplified examples show three candidate planes.

12 mm / 11.1 mm



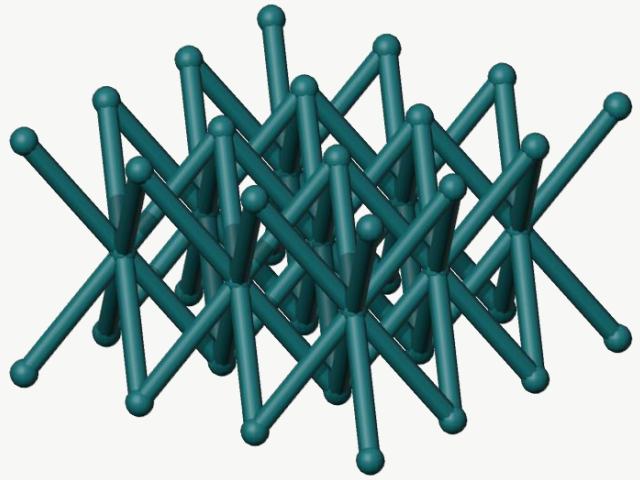
Standard pitch / rise

16 mm / 11.1 mm



44.5°: fails the angle rule

16 mm / 14 mm



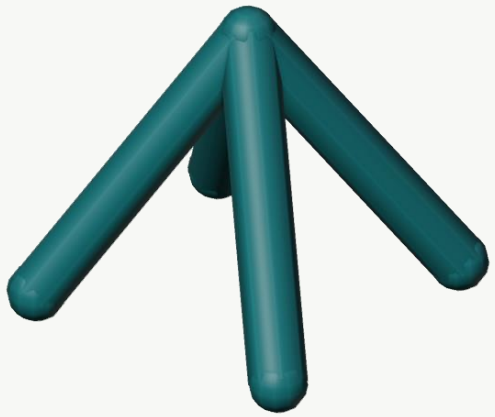
About 51.1°: eligible

Nearest run = pitch / $\sqrt{2}$. Larger rise can restore connections; local upper levels are placed separately.

Lower node diameter

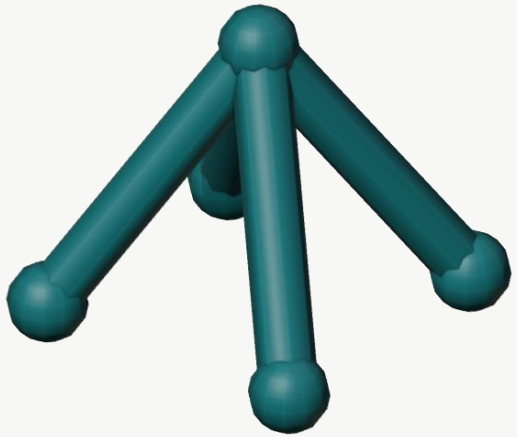
Same cell, 12 mm pitch, 11.1 mm rise and 2.4 mm struts.

2.4 mm



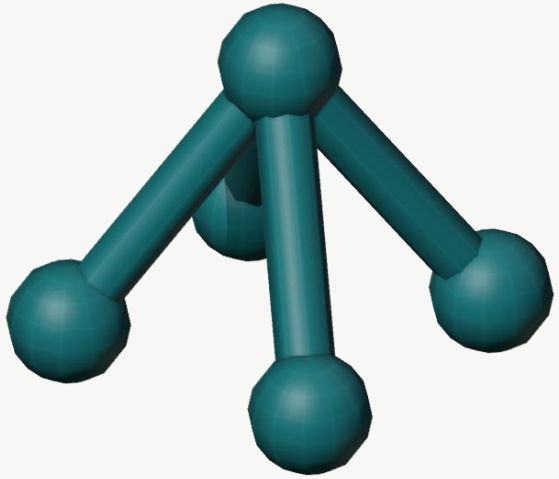
Smaller junctions

3.3 mm standard



Standard junction size

5 mm



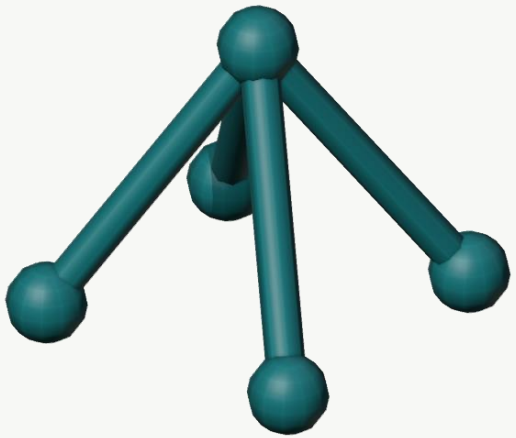
Larger junctions

Node diameter changes the junction envelope. It leaves node centers and strut centerlines in place.

Lower strut diameter

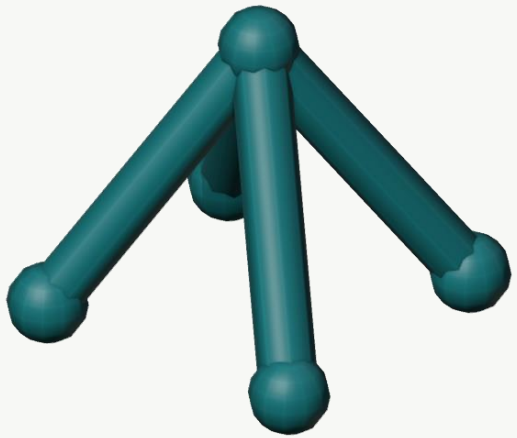
Same cell and 3.3 mm nodes. Only member thickness changes.

1.6 mm



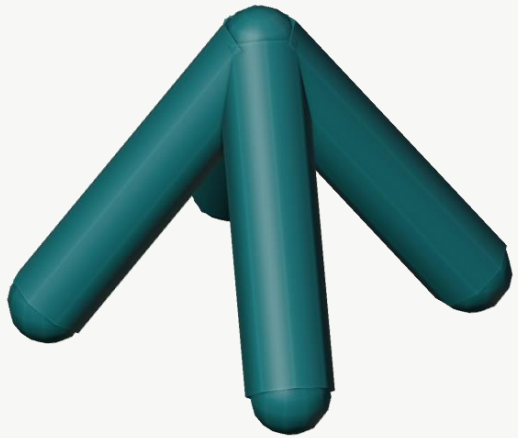
Thinner members

2.4 mm standard



Standard member thickness

3.6 mm



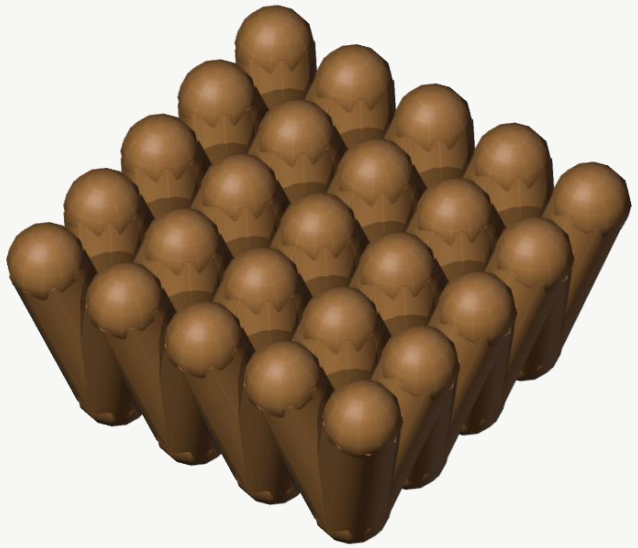
Thicker members

Strut diameter changes material around the centerline. The generator does not resize the nodes with it.

Top contact pitch

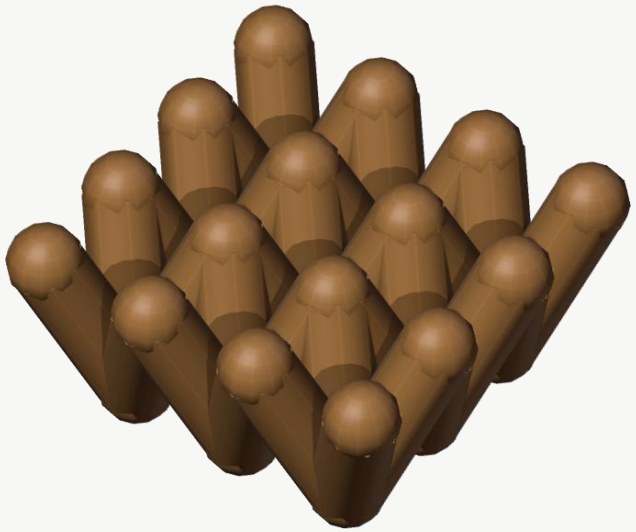
Two local upper levels use 5 mm nodes and struts. These examples illustrate a flat region.

6 mm standard



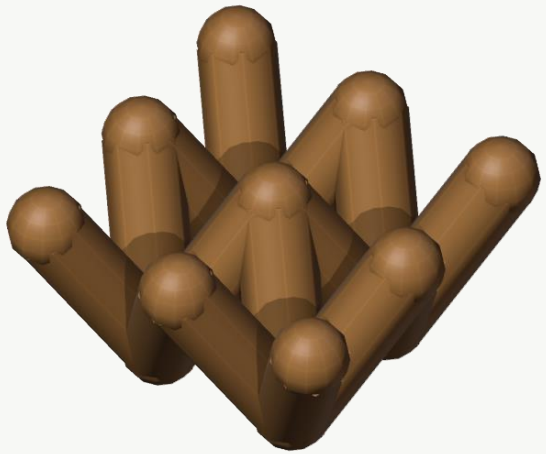
1 mm nominal node gap

8 mm



3 mm nominal node gap

10 mm



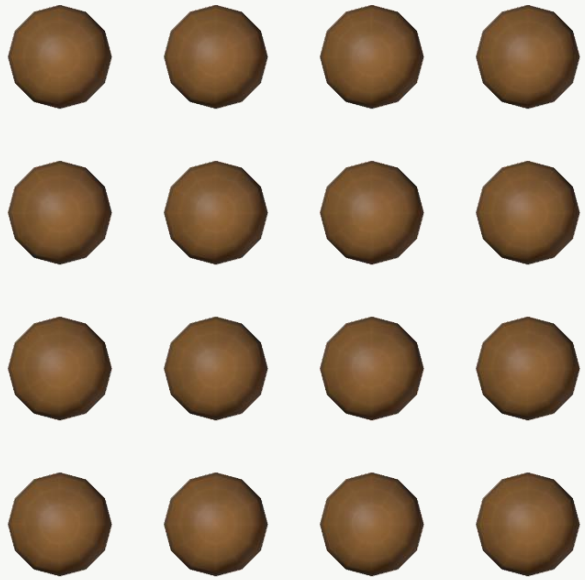
5 mm nominal node gap

Contact pitch is measured in world XY. Small or curved regions may receive additional sampled contacts.

Top node diameter and clear gap

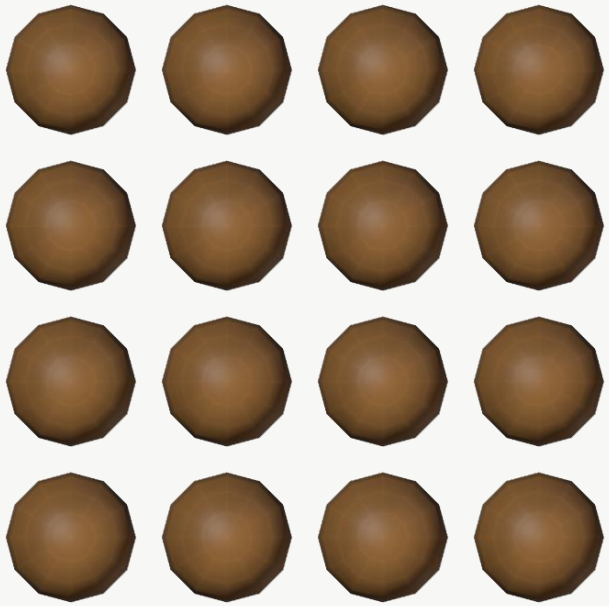
Top view at a fixed 6 mm pitch. The struts are hidden so the node spacing is visible.

4 mm nodes



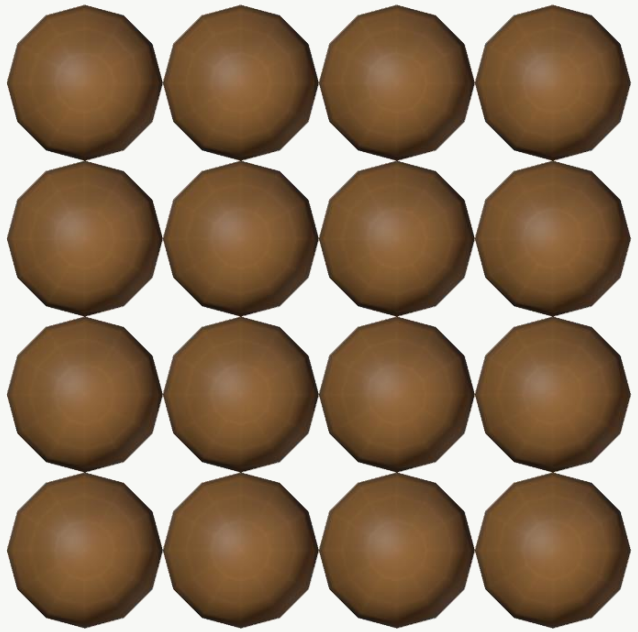
2 mm nominal gap

5 mm standard



1 mm nominal gap

6 mm nodes



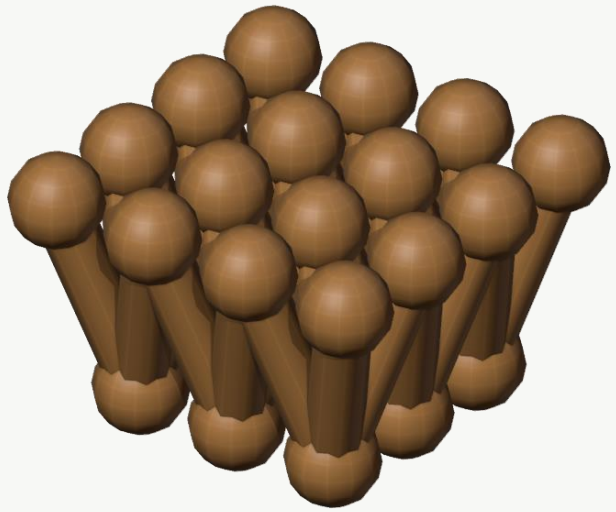
0 mm nominal gap

Nominal gap = pitch − diameter. Extra contacts and sloped surfaces can change actual local clearance.

Top and transition strut diameter

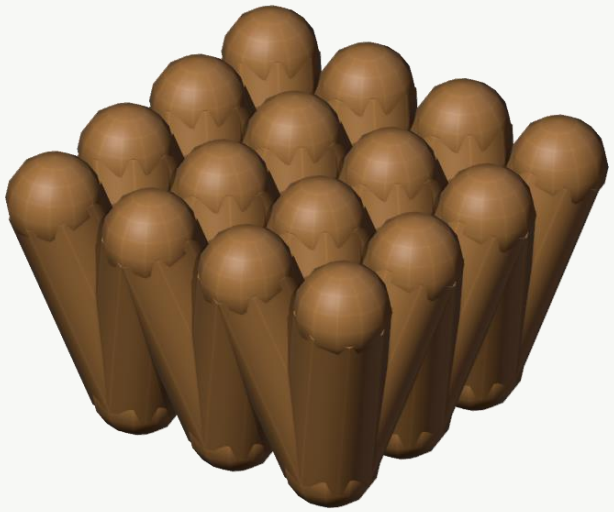
Top pitch 6 mm and node diameter 5 mm. Two upper node levels follow each selected surface.

3 mm struts



More open space between nodes

5 mm standard



Standard dense-region struts

6 mm struts



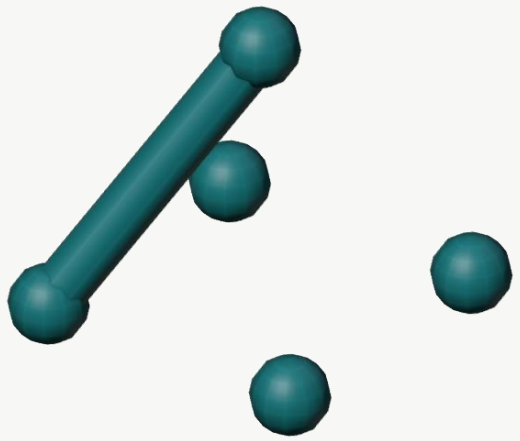
More material between nodes

Top AND transition members use this constant diameter. Transitions are not tapered and can span several layers.

Supports per node

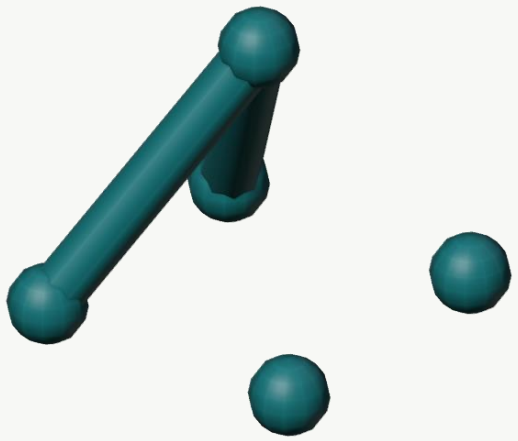
One upper node and four equally near lower candidates. Pitch and angle stay unchanged.

1 support



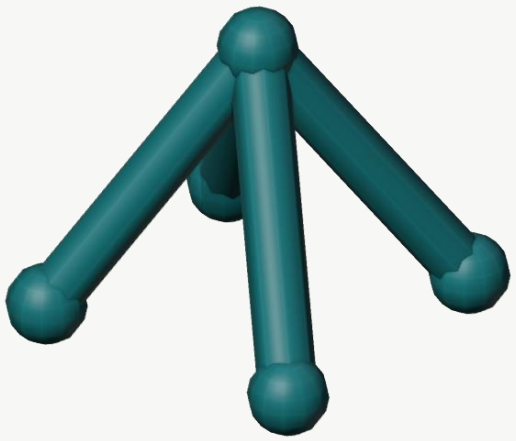
One selected connection

2 supports



Two selected connections

4 standard



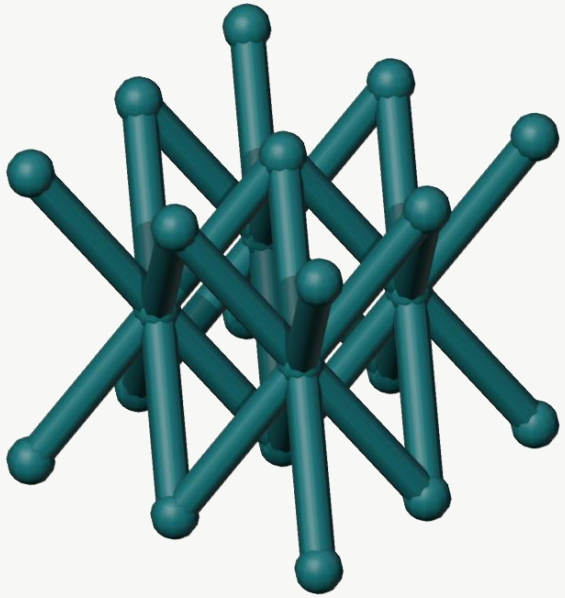
Four selected connections

Maximum selected connections per stage. Unselected candidates are shown here; unused nodes are omitted.

Minimum strut angle

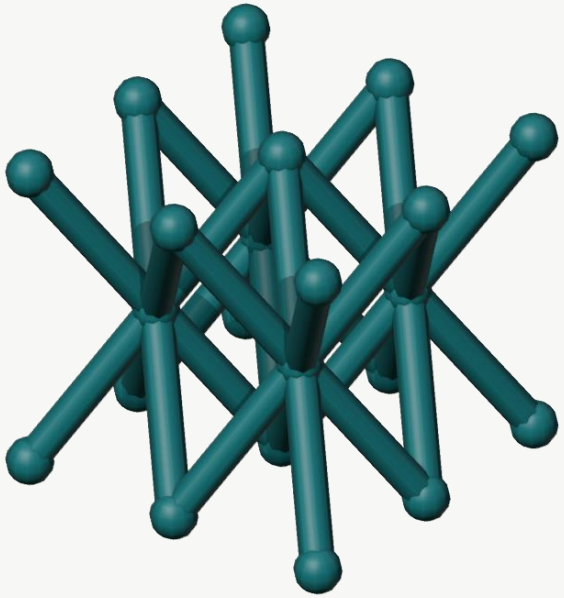
Same 12 mm pitch and 11.1 mm rise. The nearest lower connections are 52.6°.

45° threshold



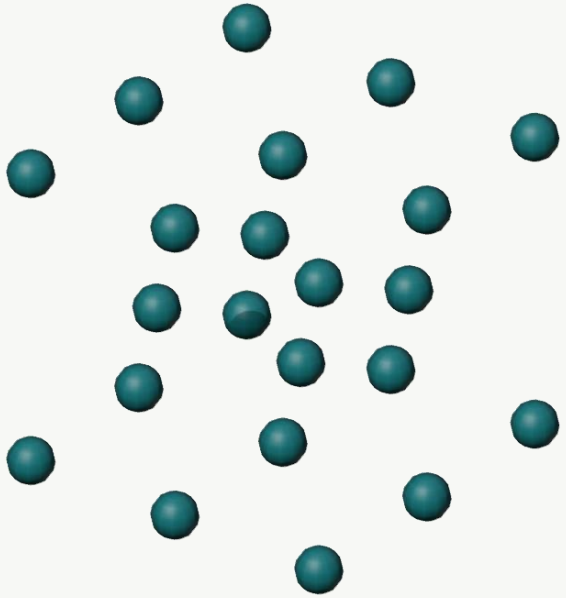
The nearest connections remain

50° standard



The nearest connections remain

55° threshold



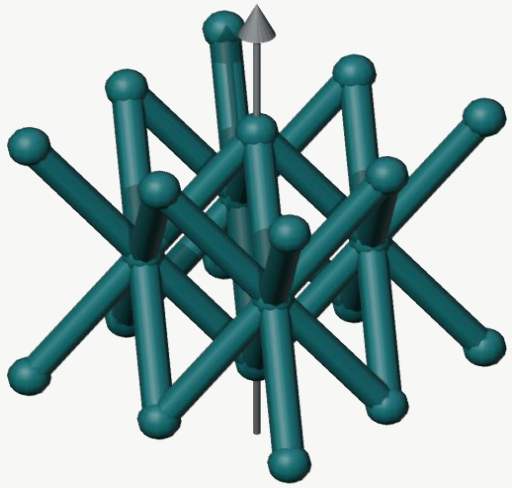
Lower connections excluded

The angle is measured above world XY. Selection adds 0.5°; dots illustrate candidates, not final isolated nodes.

Build direction and XY grid rotation

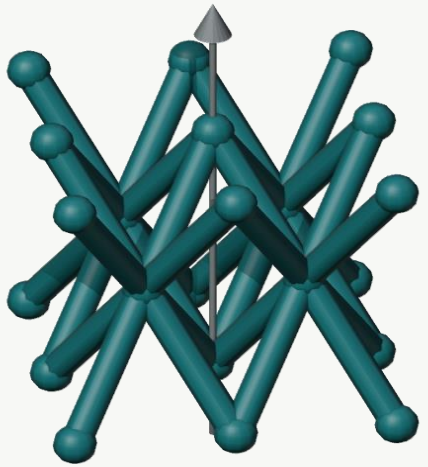
World +Z is fixed. The orientation hint rotates only the lower grid within XY.

Unrotated lower grid



x_hint = [1, 0, 0]

Lower grid rotated 45°

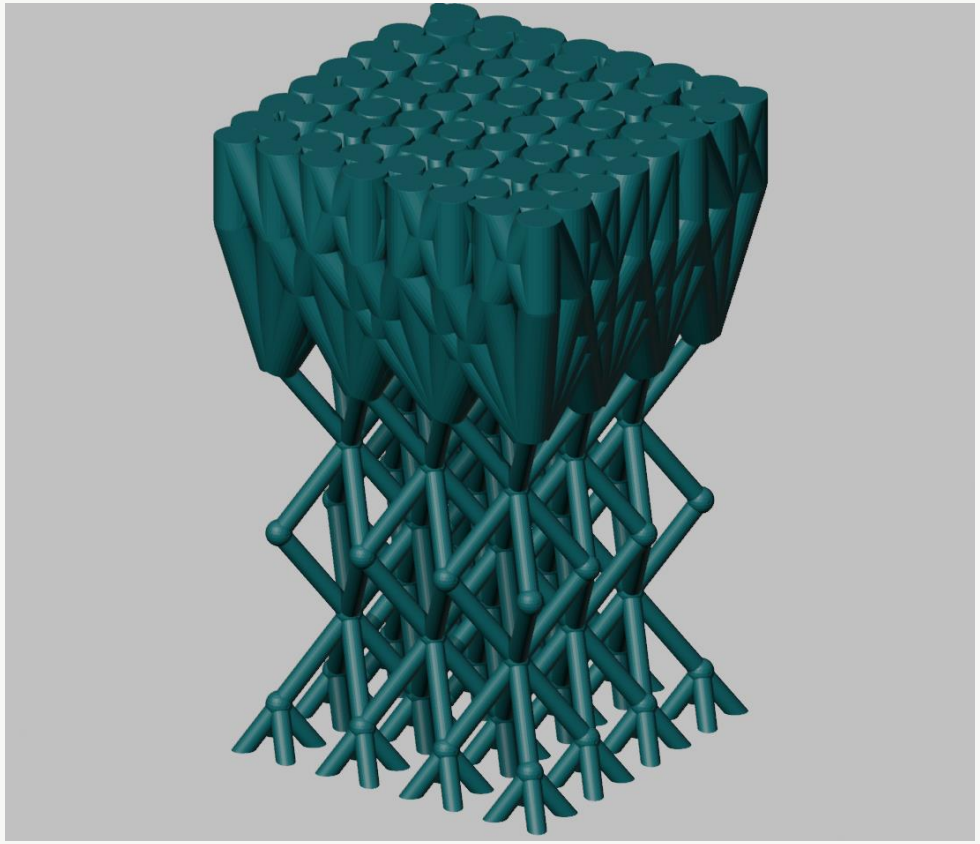


x_hint = [1, 1, 0]

Reorient the STEP body before processing if another build direction is needed. The upper contact grid stays in world XY.

Local upper layers and transition struts

Planar and curved roofs / ledges below the selection angle receive local upper members.



Node level	Z position
Upper contact level	80.0 mm
Local level below it	68.9 mm
Regular lower levels	5.55 to 72.15 mm

The local pair is 11.1 mm apart. It is added to the regular grid, giving nine distinct node heights here.

Top and transition struts: constant 5 mm diameter. Transition links can reach several lower levels.

Example: 50 × 50 × 80 mm, 0 mm cover. Floors are not selected as roofs; exactly 45° is excluded.

Geometry validation and manufacturing checks

A PASS report confirms the implemented checks; it does not certify a printing process.

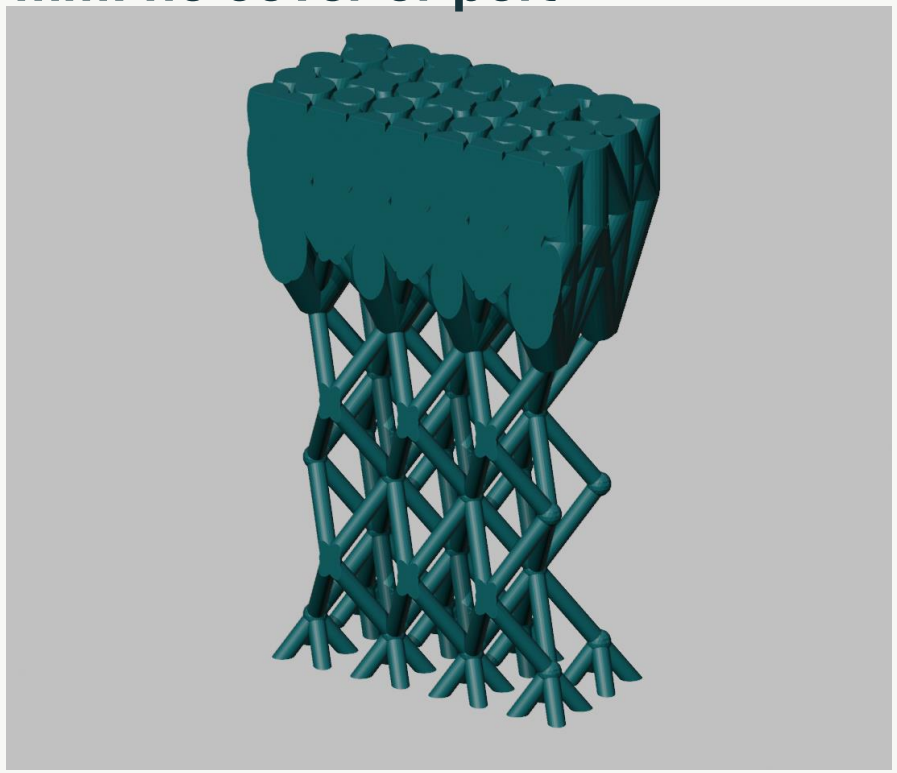
Check	Scope	What is checked
CAD containment	Both outputs	Members remain within the approved CAD boundary
Angles and connections	Both outputs	Member geometry, selected angles and node attachment
Local upper contacts	Both outputs	Finite CAD contact areas and sampled surface coverage
Powder opening	Covered mode	Whole-member exclusion and clearance at the port
Open ends and leaf branches	0 mm cover	No side cuts or dangling interior branches
Export verification	STEP / STL	CAD round-trip / mesh closure and facet composition

Still to assess: strength, global connectivity, complete powder evacuation and printing performance.

Optional cover and powder-removal opening

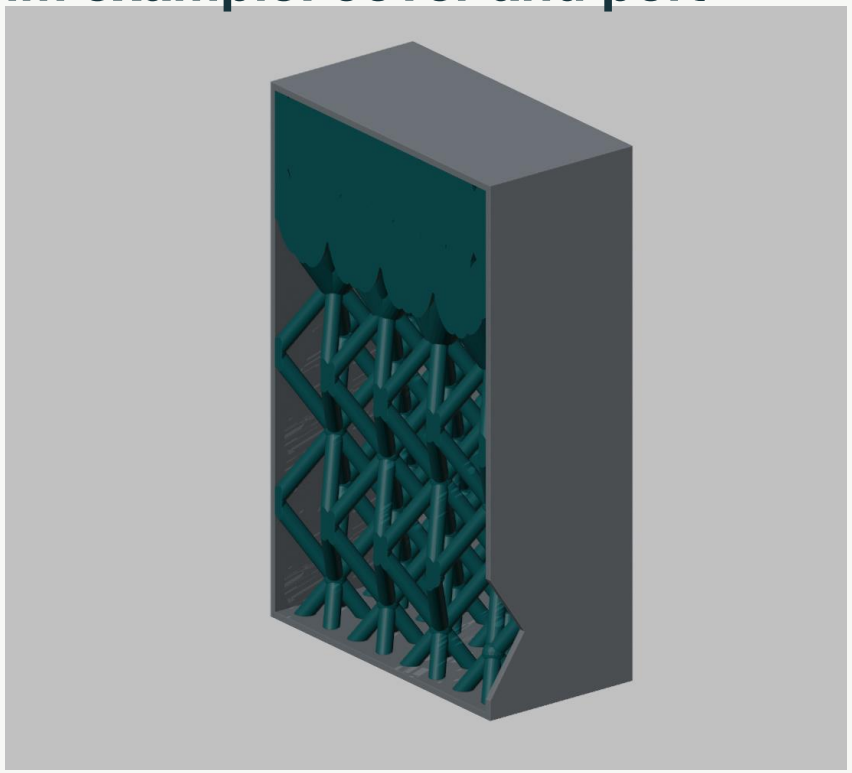
0 mm creates lattice only. Positive thickness creates an inward cover; the default is 2 mm.

0 mm: no cover or port



Open ends only at local upper / lower boundaries

1 mm example: cover and port



12 × 18 mm diamond; affected struts removed whole

At 2 mm, the same inward-cover rule leaves less interior space. Opening size and position remain editable.

STL mesh detail and STEP output

Same 5 mm CAD cylinder. Only the absolute STL tessellation deflection changes.

0.1 mm



100 triangles in this sample

0.03 mm — default



112 triangles in this sample

0.01 mm



196 triangles in this sample

Smaller deflection gives finer STL detail and usually more triangles. It does not change CAD feature dimensions.

STEP/STP exports CAD surfaces and has no mesh-deflection setting. Surface-probe sampling is separate.